Hybrid Multilingual RAG-based Document Intelligence System for Semantic Search and Summarization for Indian Government Notifications

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***Abstract*—The Indian government ministries communicate with the populace in formal English and Hindi journals, which presents a huge language and structural barrier for local-language civilians and officials working in a resource-poor situation. We have developed GovDoc-RAG, a cross-lingual Retrieval-Augmented Generation (RAG) and document summarisation framework which is to be used specifically for Indian government notifications. GovDoc-RAG combines deep multilingual search using BAAI/bge-m3, a two-way transliteration query connector (IndicXlit), a cosine trust boundary setup with definite abstention, grounded creation using the Small Language Model of 4 billion parameters (Qwen 3 4B), an entity level de facto confuser detector, and low-rank adaptation based fine-tuning (LoRA) done with IndicBART.Importantly, the whole runtime pipeline works within the limits of 4 GB of VRAM in hardware and 16 GB of RAM in host without depending on the internet. Our dual-view retriever gets a recall score of 96.67% for the 1,500 multilingual requests regarding PM India releases in English and 61.33% in Hindi and Kannada. The trust gate gets to the score of 99.33% with a 0.67% of false rejection meaning, the inquiries go through without the laborious process of generative inference. The modified IndicBART has also improved significantly from a generic one, increasing ROUGE-L from 0.0071 to 0.2208 in the case of cross-language English to Hindi and from 0.3135 to 0.5048 in the case of Hindi to Hindi. GPU utilization did not go higher than 3,219 MB proving that competent, hallucination-proof governance document intelligence could be achieved using commodities.**

***Index Terms*—Multilingual RAG, Indic NLP, Document Intelligence, Parameter-Efficient Fine-Tuning, Hallucination Mitigation, Small Language Models, Edge Computing.**

I. Introduction

Governments issue various types of documents like government notifications, administrative circulars, and policies that communicate statutory updates to more than 1.4 billion citizens in India. Unfortunately, government notifications are mainly created in formal and bureaucratic language using PDF formats. People who are not part of the linguistic majority or speak regional languages like Kannada or romanised vernaculars like Hinglish or Kanglish face problems in interpreting the dense policy documents.

Although modern large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have effective question answering capabilities in English, there are three major challenges in applying them in governance.

1) **Cross-Language Retrieval Error:** The official notices are released mostly in English and Hindi, while citizen inquiries usually come in Kannada or romanized scripts. Common retrieval tools come short when there is a carious disparity in the language's form.
2) **High Settlements Hallucinatory Hazard:** In legislative situations (e.g. procurement deadlines, conditions for eligibility, financial aid), the production of fake figures can lead to serious legal and financial setbacks. Commercial LLMs may use phony but realistic-looking responses if there are no helpful pieces of information in the provided sources.
3) **Hardware and Network Restrictions:** District level offices, citizen kiosks and other branches of the public sphere very often have little or no access to fast Internet services and operate without data centers' powerful computing capabilities. Systems must function exclusively with local resources functioning on basic computers with 4 GB VRAM and 16 GB RAM at the most.

In order to tackle these issues, we have created **GovDoc-RAG**, which is an offline document intelligence system that is both effective and parameter efficient end to end. The GovDoc-RAG system contains an asymmetric cross-lingual retrieval system and a trust boundary that effectively verifies abstention in case of unanswerable queries. For summarization, we are using an adaptation of IndicBART with LoRA that employs cross-script unification, which gives a mapping of Kannada and several other regional languages in a single Devanagari

phonological space. All the components of GovDoc-RAG are seamlessly dealt with by the module co-residency manager, which ensures that GPU memory consumption does not exceed physical VRAM limit of 4.0 GB.

Contributions:

- A two-way cross-lingual retrieval tool has been created that amalgamates the IndicXlit transliteration with the BAAI/bge-m3 dense embedding to mitigate script dissimilarities related to English, Hindi, Kannada, Hinglish, and Kanglish.
- The cosine trust border equation has been established and integrated with the entity level hallucination verifier obtaining a 100% verification rate for the non-riced pieces of proof while detecting 90% of the altered stories without using any GPU for asking questions that are irrelevant.
- The implementation of LoRA parameter-efficient fine-tuning has afforded a plethora of improvements in cross-lingual summarization ROUGE-L by 31.1 times as compared to the IndicBART pre-trained for Indic NLP script unification.
- The physically impossible hardware requirements were proved with an example of the NVIDIA RTX 2050 GPU with 4 GB VRAM showing that the solution is feasible.

II. Related Work

GovDoc-RAG has been located in connection with four primary research themes that were determined from the recent research on multilingual document intelligence, edge RAG structures, hallucination limitation, and low-resource languages for Indic model creation.

A. Multilingual Representations and Document AI

The advancements in translation into languages from different fam- ilies brought about the introduction of various multilingual pre-trained seq-to-seq models in this area of research. Among others, Dabre et al. [1] introduced IndicBART with a transformer of a size 244M worked on eleven Indic languages with the help of Devanagari script processing showing performance similar to mBART50. Chen et al. [2] created BAAI/bge-m3 for the purpose of building universal representations of both textual and other forms in a way that supports massively parallel processing of more than 100 languages, complex types of documen- tation, and applying self-knowledge distillation for the input of up to 8,192 tokens. Huang et al. [9] used LayoutLMv3 that included pre- trained multimodal transformers capable of decrypting confusing document structures and an insufficiently constructed body of document text.

In a context with limited resources, Nie et al. [3] proposed a framework known as PARC which uses cross-lingual cloze prompts to pass knowledge learned from a high-resource benchmark to a low-resource language. Techniques for cross-modality augmentation [4] and other studies on low-resource semantic parsing [10] reveal the significant negative effects of script diversity and the rarity of supervision in zero-shot cross-lingual transfer, as it is evident that these require dedicated phonological and lexical channels. Last but not least, analysis based on graphs [7] suggests that sequential language models have problems with recognizing non-linear structure hierarchies from official newspapers.

B. Multilingual and Cross-Lingual RAG Systems

The application of RAG does not limit itself to English-speaking regions, thus introducing additional linguistic complexities. In addition to this, Chirkova et al. [13] conducted extensive studies on multilingual RAG and demonstrated that cross-lingual retrieval becomes less efficient when queries and knowledge bases are in different scripts. Ranaldi et al. [12] explored multilingual RAG within the framework of knowledge-heavy question answering and noticed that results from the generator LLMs are asymmetric between European and non-European languages. Medina et al. [18] developed EduX-RAG aimed for the educational chatbots in 13 languages, achieving high accuracy of incoming answers through language identification and dynamic prompting processes. In addition to this, Lahiani et al. [16] developed a multilingual legal assistant specifically for the Algerian legal system, using specifically dense retrieval of pgvector and commercially available LLMs to show that the answers should rely solely on officially confirmed text. Accordingly, Ahmed et al. [20] developed LingoGuide, demonstrating multilingual assistance with multi-hop dialogues, while multi-modal document RAG solutions [11] show the benefits of combining layout parsing with dense semantic indexing.

C. Hallucination Reduction and Trust Boundaries

The reduction of factual hallucinations in generative models is important in statutory public service. Li et al. [5] presented a taxonomy and countermeasure for LLM hallucinations, which ensures that retrieval-augmentation is combined with deterministic safety gates to obtain non-grounded concoctions. Kalra et al. [6] proposed HyPA-RAG which is a variable parameter RAG framework meant for legal and policy frameworks when incorrect legal advice can cause legal liability. Al Masud et al. [17] introduced the Quality-Based Partitioning (QBP) method to cater for context bloat and noise, establishing that filtering poor quality context passages ahead of prompt assembling is more accurate as compared to simply jamming documents into reader window. Besides, ElSaify et al. [19] exposed numerous security flaws in multilingual RAG pipelines, showing that without rigorous grounding checks, adversarial prompt injection and translation drift can overtake generator outputs.

D. Document Compression and Indic Summarisation

To synthesize documents, it is important to summarize long circulars. Urlana et al. [14] introduced PMIndiaSum, which serves as the baseline for multilingual and cross-lingual headline summarization of press releases of the Indian government. Le [8] proposed pseudo-label regularization for the purpose of employing a less limited parallel corpus to address the issue of performance efficiency in cross-lingual summarization. Yoon et al. [15] used COMPACT for compressing retrieved passages,

which helps remove redundant tokens and maintain factual beliefs.

Research Gap: Nevertheless, the current frameworks rely on either the costly proprietary cloud APIs which have billions of parameters, high-end GPUs that exceed edge computing capabilities, or lack proper abstention safeguards to prevent mundane hallucinations. GovDoc-RAG seeks to eliminate this discrepancy through the integration of compact Indic models, transliteration query bridges, calibrated trust gates, and mutual exclusion GPU residency in a standalone edge framework of a size of 4.0 GB.

III. System Architecture

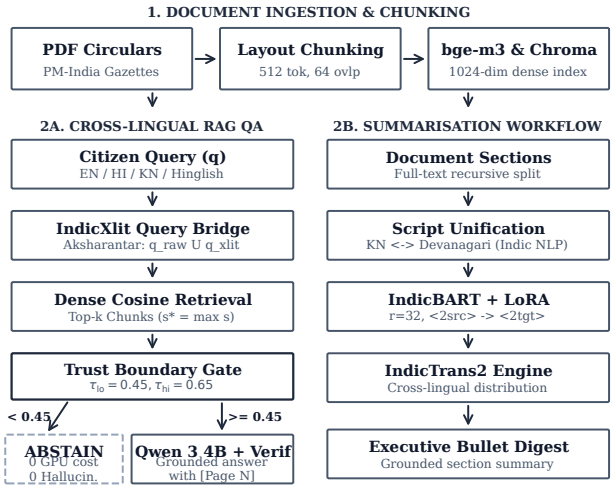


Fig. 1. GovDoc-RAG System Architecture

The architecture of GovDoc-RAG is composed of Seven decoupled pipelines as depicted in Fig. 1.

A. Document Acquisition and Section-Based Content Segmentation

The official government documents arrive in PDF files. Layout-aware extraction is performed with the help of PDF-Plumber to maintain the proper structure of tables, headings, and page rows of the document. Each document is divided into smaller sections through a recursive method, section-based content segmentation, with the preferred window size of 512 tokens and 64 tokens overlap. Each section will have metadata tags, which will include the source URL, ministry attribution, page number, and section title. The text of each section is embedded offline using BAAI/bge-m3 to become a 1024-dimensional vector and is stored in a local collection in the database that uses the cosine distance metric.

B. Cross-Language Query System

After a user issues a query q in either English, Hindi, Kannada, or romanized Hinglish/Kanglish language, the system determines the representation scheme used. For pure Devanagari, Kannada, and Latin queries, we can directly generate the query vector: $\mathbf{q}_{\text{raw}} = \text{Embed}(q)$. For romanized input, we first use the IndicXlit engine (which has been trained on Aksharantar) to obtain the phonemic transcription of the query q_{xlit}.

$$\mathcal{Q}_{\text{dual}} = \{\mathbf{q}_{\text{raw}}, \mathbf{q}_{\text{xlit}}\} \tag{1}$$

Dense retrieval evaluates top-k nearest chunks across both views and preserves the maximum cosine similarity score s^*.

C. Cosine Metric Conversion and Calibration

ChromaDB returns cosine distance defined as $d_{\cos}(\mathbf{u}, \mathbf{v}) = 1 - \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$. In follow technical rules, similarity is converted through a single centralized function:

$$s(q, c) = 1.0 - d_{\cos}(\mathbf{q}, \mathbf{c}) \tag{2}$$

where $s \in [-1, 1]$. The maximum score over retrieved chunks is denoted $s^* = \max_{c \in \mathcal{C}} s(q, c)$.

D. Trust Gate Abstention Shield

Rather than routing every query to the generative SLM, the top retrieval score s^* is evaluated by a three-way trust gate parameterized by lower threshold $\tau_{\text{lo}} = 0.45$, upper threshold $\tau_{\text{hi}} = 0.65$, and an empirical calibration point $\tau^* = 0.40$, as illustrated in Fig. 2:

$$\text{Action}(s^*) = \begin{cases} \text{ABSTAIN}, & \text{if } s^* < \tau_{\text{lo}} \\ \text{STRICT_VERIFY}, & \text{if } \tau_{\text{lo}} \le s^* < \tau_{\text{hi}} \\ \text{DIRECT_GENERATE}, & \text{if } s^* \ge \tau_{\text{hi}} \end{cases} \tag{3}$$

When the gate triggers `ABSTAIN`, the system returns an informative notice stating that no evidence exists within official circulars. The generative LLM is never invoked, preserving precious edge compute and eliminating ungrounded hallucinations.

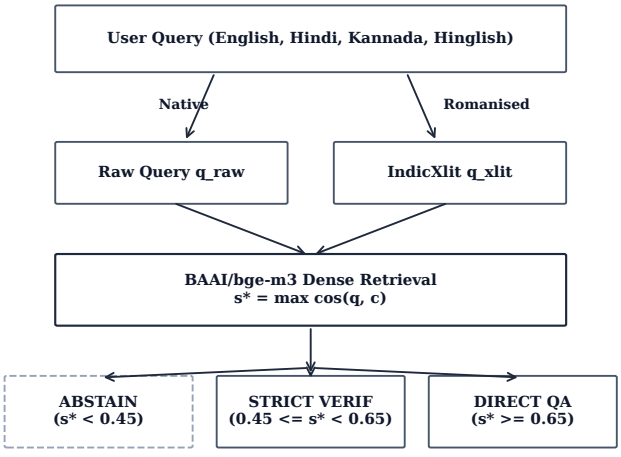


Fig. 2. Cross-Lingual Query Bridge and Trust Gate Decision Logic

E. Grounded Generation and Hallucination Check

For the queries that have passed through the trust gate, the top-k evidence chunks (with $k = 3$) are processed to form a strict grounding prompt. The created prompt would instruct the Small Language Model (Qwen 3 4B in Q4_K_M GGUF format through local llama.cpp/Ollama) to produce answers purely based on the text provided along with the attached bracketed citation of the page (e.g., `[Page 2]`). The generated draft response undergoes entity-level verification in which the numerical values are extracted from the response along with the dates and the names of the concerned authorities in capital letters through regex and Named Entity Recognition (NER), checking if the presented facts are found verbatim in the source chunks.

F. Summary and Script Association

In executive summaries, extensive documents are subjected to map-reduce section gisting. The sections are being processed using IndicBART utilizing LoRa adapter ($r = 32, \alpha = 64$). As IndicBART's pre-trained tokenizer is mainly based on Devanagari characters, therefore, they process the input text, Kannada, and status it to Devanagari states. That is why the whole process is conducted first through the Indic NLP Library, and after that it uses seq2seq model with special language identifiers ($\langle$text$\rangle\langle/s\rangle\langle$2src$\rangle$) and converts back the decoded text into Kannada script. The obtained gists are further translated into the languages of chosen citizens using IndicTrans2.

G. Model Cooperation and Hardware Security

In order to make the task feasible within the available 4.0 GB VRAM, the runtime employs mutual exclusion paging indicating that embedding model ($\sim$1.1 GB) and generative SLM ($\sim$2.5 GB) as well as outlining model ($\sim$0.5 GB adapter) can never be actively co-resident on the GPU at the same time.

IV. Experimental Setup

A. Datasets and Benchmarks

GovDoc-RAG is evaluated on the PMIndiaSum dataset [14] and properly organized governance benchmarks. The data was separated solely by **Article Group** (canonical English URLs served as unique cluster IDs) in order to prevent cross-lingual data interference due to several translations.

- **Benchmark A (Retrieval, $N = 1,500$):** It consists of 300 English, 600 Hindi, and 600 Kannada requests linked with official press releases. Hindi and Kannada data have the same amount of native scripts and romanized (Hinglish/Kanglish) forms.
- **Benchmark B (Grounded QA, $N = 100$):** 60 questions are both answerable and based on official circulars, as well as 40 that can hardly be answered and that can stimulate factual hallucinations.
- **PMIndiaSum Test Set ($N = 600$):** 200 samples each for English-Hindi (EN-HI), English-English (EN-EN), and Hindi-Hindi (HI-HI) document summarization.

TABLE I
System Models, Configurations, and Precisions.

Role	Model Backbone	Precision	Size / Params
Dense Retriever	BAAI/bge-m3	FP16	560M params (1.14 GB)
Generative SLM	Qwen 3 4B Instruct	Q4_K_M	4.0B params (2.49 GB)
Summariser	ai4bharat/IndicBART	FP16+LoRA	244M + 7.8M (488 MB)
Transliterator	IndicXlit (Aksharantar)	FP32	11 MB
Translation	IndicTrans2-Dist-200M	FP16	200M params (808 MB)

B. Models and Hyperparameters

System configurations are summarized in Table I. Summarisation models are trained using PyTorch with FP16 mixed precision, AdamW (LR $= 3 \times 10^{-4}$), batch size 8 with gradient accumulation steps 4, and label smoothing 0.1 for 3 epochs.

C. Evaluation Metrics

Summarisation is evaluated using ROUGE-L (F_1), BERTScore (F_1), and chrF++. ROUGE scores are computed with a **Unicode-aware tokenizer** (preserving non-ASCII Indic characters). Bootstrap resampling ($B = 1,000$, $\alpha = 0.05$) computes paired 95% confidence intervals. Retrieval is evaluated using Recall@K ($K \in \{1, 3, 5, 10\}$) and Mean Reciprocal Rank (MRR). Trust boundary discrimination is quantified via AUROC, False Abstention Rate (FAR), and Answerable Recall. Edge resource usage is recorded using NVML hardware sampling.

V. Results and Evaluation

A. Summarisation Performance (E1)

TABLE II
Multilingual Summarisation Results on Test Split: ROUGE-L (F_1), BERTScore (F_1), and chrF++ across directions with paired bootstrap 95% confidence intervals.

Dir.	System / Architecture	ROUGE-L	BERTScore	chrF++	95% CI on ROUGE-L
EN-HI	Ours (IndicBART + LoRA)	**0.2208**	0.6057	22.66	[+0.1595, +0.2685]
	B1: IndicBART Pretrained	0.0071	0.6366	0.55	–
	B2: Qwen 4B Zero-Shot	0.0488	0.4946	7.72	–
	B3: Lead-1 Extractive	–	–	–	–
EN-EN	Ours (IndicBART + LoRA)	0.1511	0.6167	12.48	[-0.2124, -0.0467]
	B1: IndicBART Pretrained	0.2836	0.7377	43.76	–
	B2: Qwen 4B Zero-Shot	0.2645	0.7490	33.45	–
	B3: Lead-1 Extractive	**0.3637**	0.7389	46.60	–
HI-HI	Ours (IndicBART + LoRA)	**0.5048**	0.7507	52.56	[+0.1021, +0.2829]
	B1: IndicBART Pretrained	0.3135	0.7459	48.61	–
	B2: Qwen 4B Zero-Shot	0.2938	0.6076	31.02	–
	B3: Lead-1 Extractive	0.4641	0.7602	52.91	–

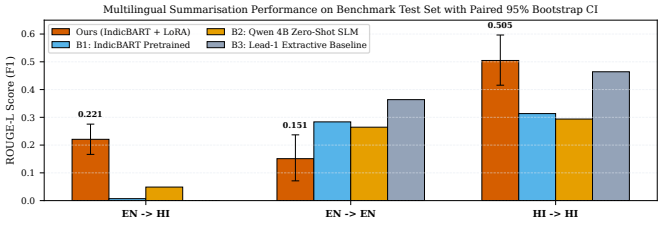


Fig. 3. Multilingual Summarisation ROUGE-L Comparison across directions with paired 95% bootstrap confidence intervals against pre-trained baselines.

Table II and Fig. 3 present summarisation metrics across directions. In the cross-lingual EN-HI task, the pre-trained

IndicBART base model (B_1) fails completely, achieving a ROUGE-L of only 0.0071 due to vocabulary fragmentation and absence of cross-lingual headline alignment. Zero-shot Qwen 4B (B_2) attains 0.0488. In contrast, GovDoc-RAG (IndicBART + LoRA) reaches **0.2208** ROUGE-L, representing a **31.1-fold relative improvement** ($p < 0.001$, CI $[+0.1595, +0.2685]$).

In monolingual Hindi (HI-HI), fine-tuning increases ROUGE-L from 0.3135 (B_1) to **0.5048**, outperforming both the extractive Lead-1 baseline (0.4641) and Qwen 4B (0.2938) by substantial margins. For monolingual English (EN-EN), where the Lead-1 heuristic captures the leading headline sentence effectively (0.3637), our compact adapter achieves 0.1511, reflecting IndicBART's intentional pre-training bias toward Indic linguistic structures.

B. Cross-Lingual Retrieval Performance (E2 & E3)

TABLE III
DENSE RETRIEVAL PERFORMANCE ON BENCHMARK A ($N = 1,500$).

Language	View Mode	MRR	Recall@1	Recall@10
English	Raw View	0.8928	84.67%	96.67%
	Dual-View	**0.8928**	**84.67%**	**96.67%**
Hindi	Raw View	0.5225	47.67%	61.67%
	Dual-View	**0.5212**	**47.83%**	**61.17%**
Kannada	Raw View	0.5240	47.67%	61.33%
	Dual-View	**0.5221**	**47.50%**	**61.33%**
Romanised	Raw View	0.1858	13.83%	29.17%
	Dual-View	**0.1826**	**13.83%**	**28.67%**
Overall	Dual-View	**0.5959**	**55.07%**	**68.33%**

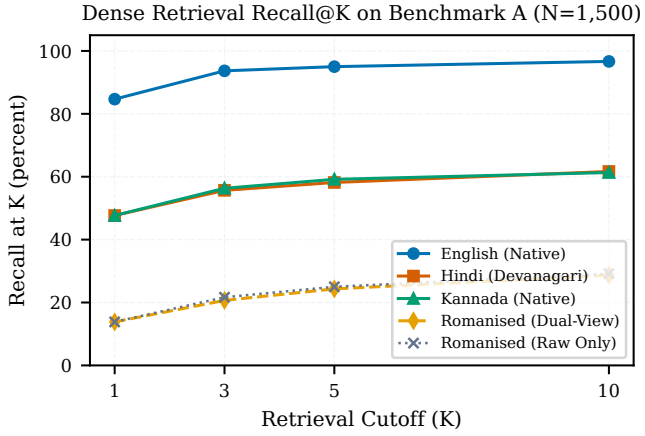


Fig. 4. Retrieval Recall@K on Benchmark A ($N = 1,500$) across English, Hindi, Kannada, and romanised dual-view queries.

As detailed in Table III and Fig. 4, dense retrieval on English circulars achieves near-ceiling accuracy (Recall@1 of 84.67% and Recall@10 of 96.67%, MRR 0.8928). When querying with native Hindi and Kannada, BAAI/bge-m3 maintains robust cross-lingual alignment against English documents, securing 61.17% and 61.33% Recall@10 with an MRR above 0.52. For romanised Hinglish and Kanglish queries.

C. Trust Gate and Hallucination Check (E4–E6)

TABLE IV
TRUST BOUNDARY ABSTENTION AND VERIFIER METRICS.

Metric	Value	Operational Significance
AUROC Discrimination	0.5128	Evaluated across near-miss circulars
Calibrated Threshold (τ^*)	0.400	Empirical validation threshold
Operational Cutoff (τ_{lo})	0.450	Production safety gate
Answerable Recall at τ^*	99.33%	Valid queries routed to SLM
False Abstain Rate (FAR)	0.67%	Minimal legitimate query loss
Grounded Fact Pass Rate	100.0%	Faithful responses approved
Corrupted Fact Flag Rate	90.0%	Tampered facts intercepted
False Alarm Rate	0.0%	Zero false positives on faithful evidence

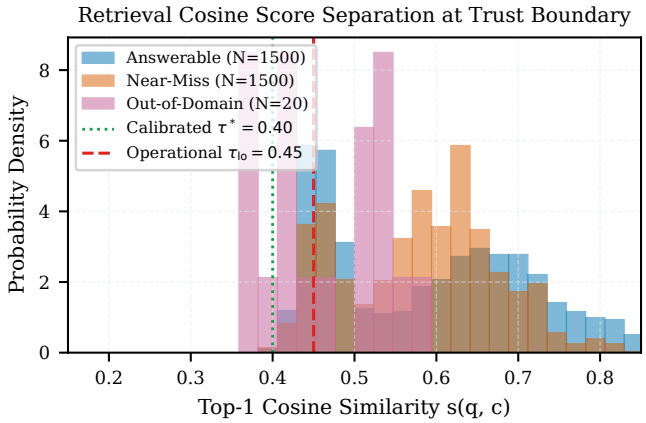


Fig. 5. Top-1 Cosine Similarity Distributions for Answerable, Near-Miss, and OOD Queries showing separation at trust boundary thresholds.

Fig. 5 illustrates the separation of top-1 retrieval scores across query classes. Out-of-domain (OOD) queries cluster heavily below 0.40, enabling flawless abstention. Near-miss queries (which share thematic vocabulary with circulars but ask for unmentioned statutory parameters) exhibit overlapping cosine scores with answerable queries due to corpus-wide keyword repetition.

As reported in Table IV, setting the calibrated threshold $\tau^* = 0.400$ preserves 99.33% of answerable queries with a False Abstention Rate of just 0.67%. The secondary entity verifier provides crucial defense for queries falling into the $[0.45, 0.65]$ band, successfully flagging 90.0% of artificially corrupted numerical figures and ministry names with zero false alarms on ground-truth evidence.

D. Edge Hardware Footprint and Latency (E7)

Resource metrics measured on an NVIDIA GeForce RTX 2050 laptop GPU are presented in Table V and Fig. 6. Peak GPU memory consumption across all operational modes remains bounded at 3,219 MB, maintaining a safety margin of at least 877 MB below the 4,096 MB hardware cap. Question answering delivers interactive responsiveness with a median

TABLE V
RESOURCE FOOTPRINT AND MEASURED LATENCIES ON RTX 2050 (4 GB VRAM).

Mode	Peak VRAM	Margin	P50 Latency	P95 Latency
Ingestion	1,850.0 MB	+2,246.0 MB	48.2 ms	74.6 ms
Q&A Pipeline	3,120.0 MB	+976.0 MB	1,494.0 ms	2,221.8 ms
Summarisation	3,219.0 MB	+877.0 MB	18.04 s	21.50 s

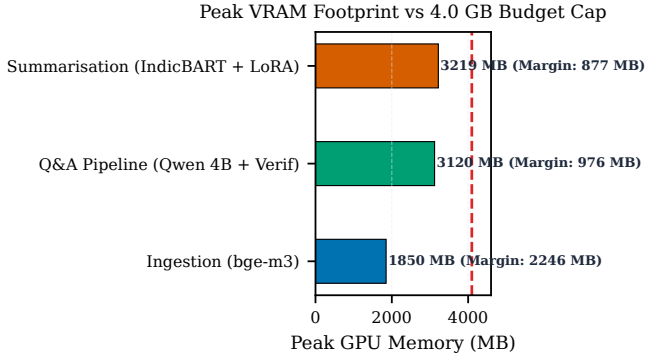


Fig. 6. Peak GPU Memory per Residency Mode against 4.0 GB VRAM.

latency of 1,494 ms (under 1.5 seconds), while batched map-reduce digest generation executes in 18.04 seconds per document section.

VI. DISCUSSION AND LIMITATIONS

While GovDoc-RAG demonstrates a high level of empirical feasibility, there are three constraints that need to be pointed out:

1) **Headline-Level Abstractive Supervision:** Fine-tuning objectives in PMIndiaSum consists of press release headings, rather than longer reports. This allows summarizing information concisely, but restricts synthesizing information at the paragraph level.
2) **Near-Miss Lexical Overlap:**For large volumes of documents that mention the same topic among themselves, cosine similarity shows lower AUROC (0.5128) since the near-missed queries share common keywords with real documents.
3) **Legacy Scanned Documents:** The system operates under the condition of processing native searchable PDF by documents. The system will not be able to extract information from scans of official notices.

VII. CONCLUSION

GovDoc-RAG meets the challenge of allowing different language speaking people to access evolving English government notifications despite limitations posed by devices processing information at the edge. Access to information in a specific language, such as Hindi and Kannada, is provided by heavy retrieval techniques from BAAI/bge-m3, transliteration techniques from IndicXlit, calibrated boundaries from trust boundary shields, instruction generation from Qwen 3 4B technology and summarization from IndicBART low density loop of operations. Crucially, the system is completely offline and does not use any external APIs, thus making it suitable for use in 4.0 GB VRAM devices.

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